Wazuh-SIEM-Deployment

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Executive Summary

This project implemented a practical **Wazuh SIEM monitoring environment** using Kali Linux as the monitored endpoint and Wazuh OVA as the central security manager. The environment was configured to collect authentication logs, monitor file integrity, and detect security-related activity.

The setup was validated through real security tests, including **failed SSH authentication, file integrity changes, authentication failures, and Nmap network scanning**. These activities successfully generated security events visible in Wazuh.

A custom Wazuh rule was also created to detect Nmap activity. The rule generated a **Level 10 alert** and was mapped to **MITRE ATT&CK T1046 – Network Service Discovery**, demonstrating the ability to extend Wazuh's detection capabilities for specific threats.

Overall, the project demonstrates a complete security monitoring workflow: **collect → detect → alert → verify**.

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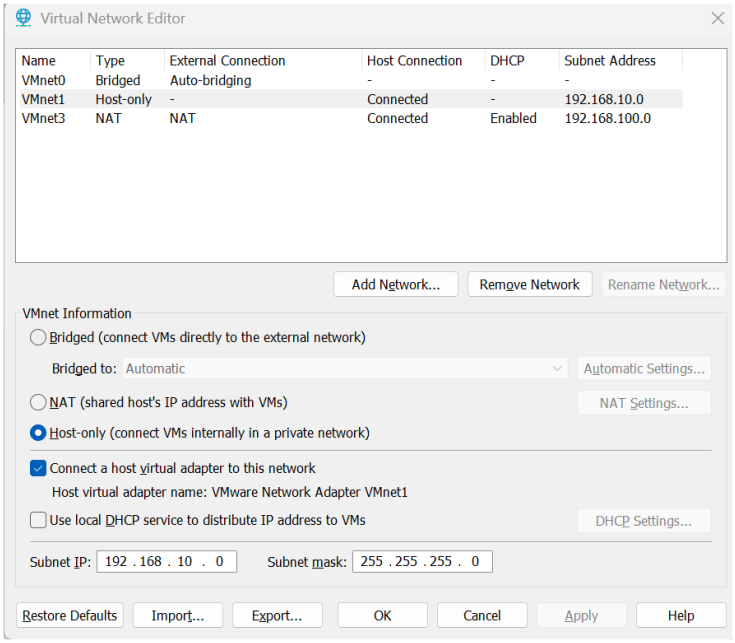
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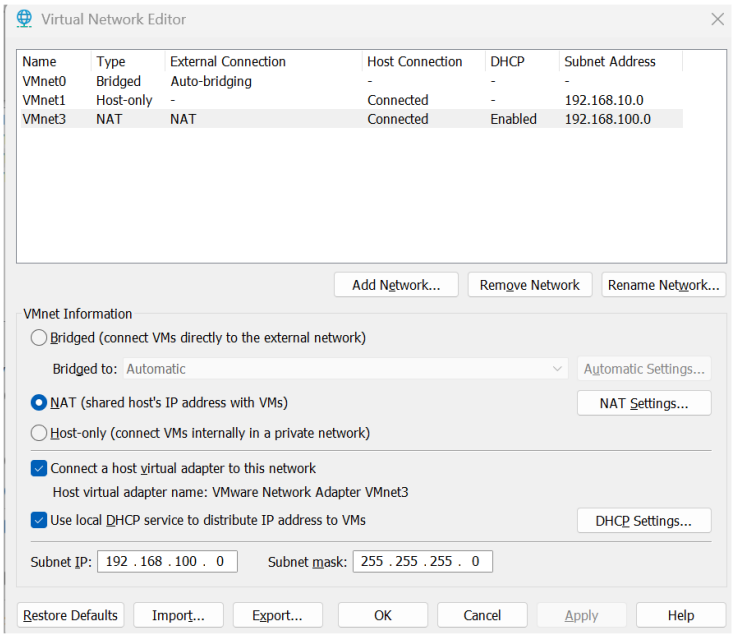
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Virtual Network Editor Setting

The VMware Virtual Network Editor is used to configure the virtual networks (VMnets) used by the lab VMs.

- **VMnet1** — Host-only network, Subnet: 192.168.10.0, connected to host, no DHCP.
- **VMnet3** — NAT network, Subnet: 192.168.100.0, DHCP enabled, connected to host.





Network Configuration

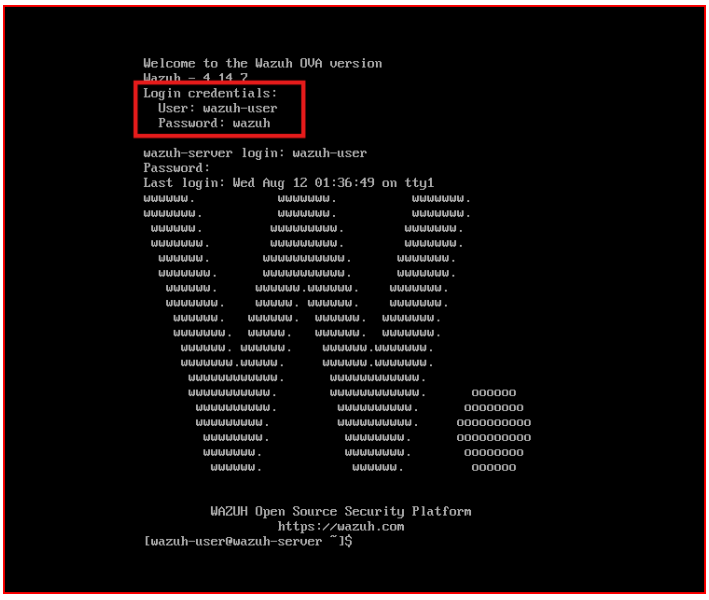
Put **all VMs** on the same network (**VMnet3**).

Machine	Recommended Static IP	Network
Kali Linux	192.168.100.10	VMnet3
Wazuh OVA	192.168.100.20	VMnet3

Note: Gateway-`192.168.100.2`

Wazuh OVA Configuration

Login with: wazuh-user / wazuh



Static IP on Wazuh OVA (192.168.100.20)

Step 1: Switch to root user

```
sudo -i
```



Explanation: Gives full administrative privileges to edit system network files.

Step 2: Create a static network configuration file

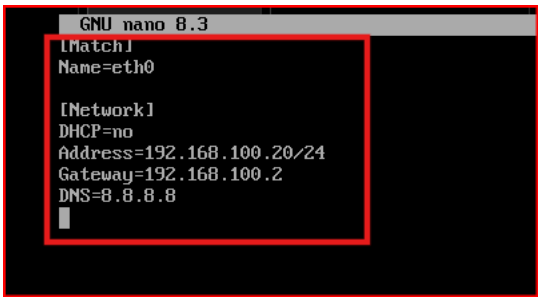
```
nano /etc/systemd/network/00-static.network
```



Explanation: Creates a new configuration file for the static IP settings.

Note: Wazuh OVA is based on **Amazon** Linux **2023**, which does not use netplan. It uses **systemd-networkd** for network configuration.

Step 3: Add the following content in the file



Explanation:

- Name=eth0 → Selects the correct network interface
- DHCP=no → Disables automatic IP assignment
- Address → Sets the static IP
- Gateway → Sets the default gateway
- DNS → Sets Google DNS server

Save the file: Press Ctrl + O → Enter → Ctrl + X

Step 4: Restart the network service

```
sudo systemctl restart systemd-networkd
```

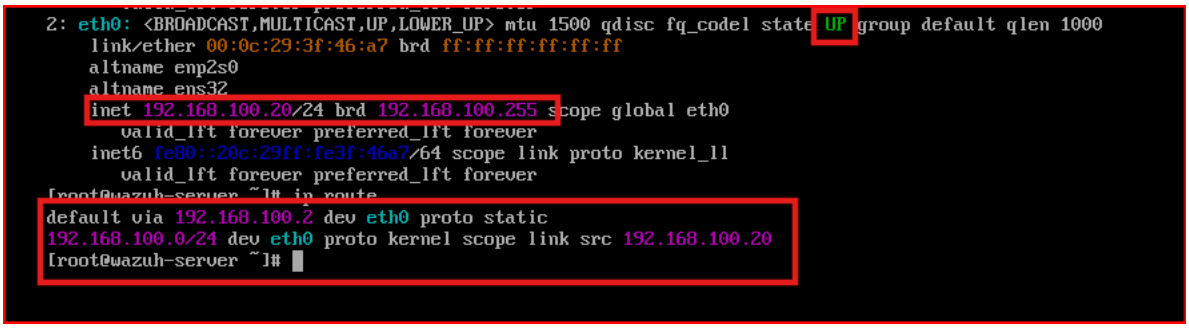


Explanation: Applies the new static IP configuration.

Step 5: Verify the IP address & default gateway

```
ip a
```

```
ip route
```

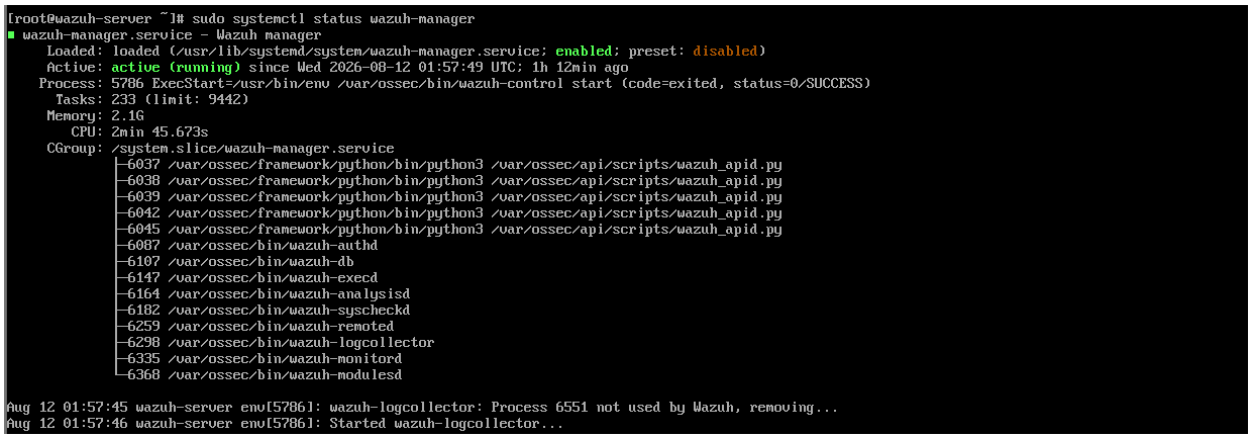


Result: Wazuh VM now has a permanent static IP: **192.168.100.20**

Check Services in Wazuh OVA

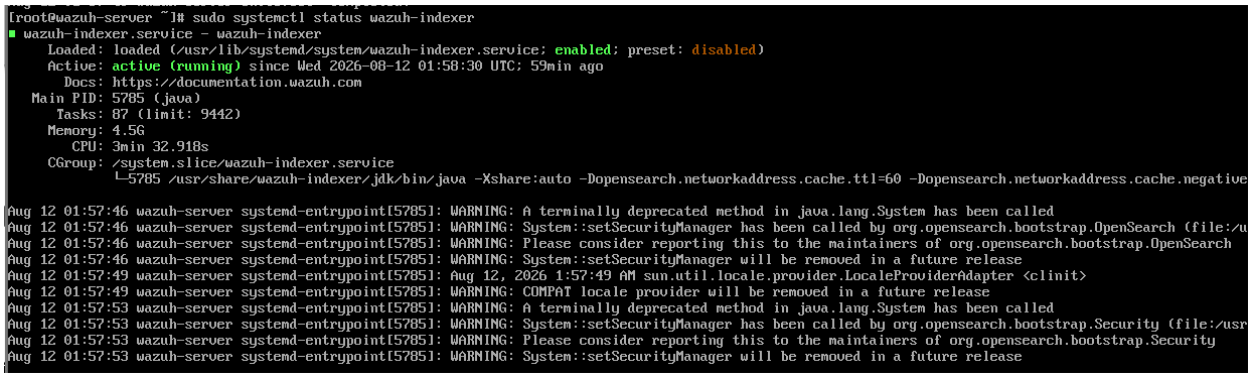
Step 1: Check Wazuh Manager

```
sudo systemctl status wazuh-manager
```



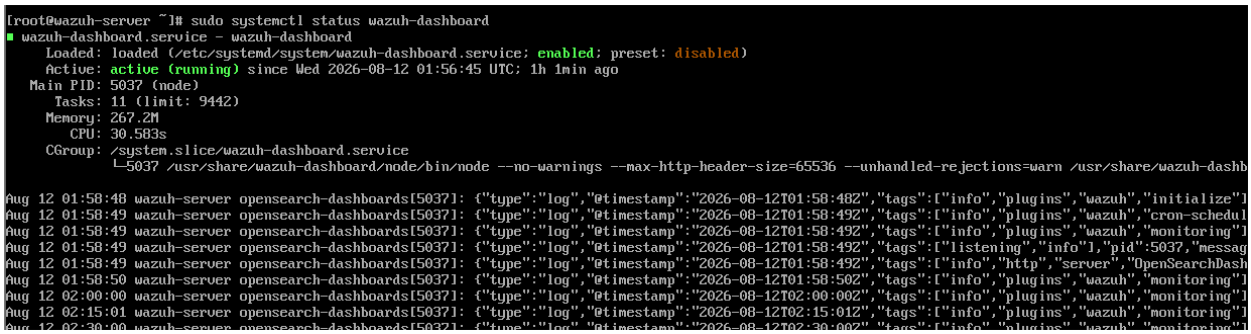
Step 2: Check Wazuh Indexer

```
sudo systemctl status wazuh-indexer
```



Step 3: Check Wazuh Dashboard

```
sudo systemctl status wazuh-dashboard
```

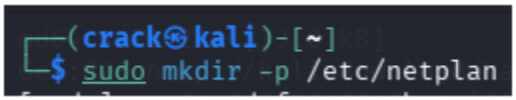


Kali Linux Configuration

Static IP on Kali Linux (192.168.100.10)

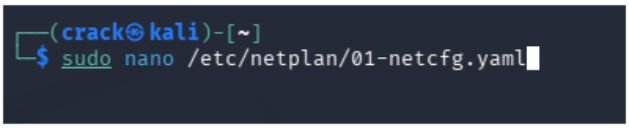
Step 2: Create the netplan directory (if doesn't exist)

```
sudo mkdir -p /etc/netplan
```



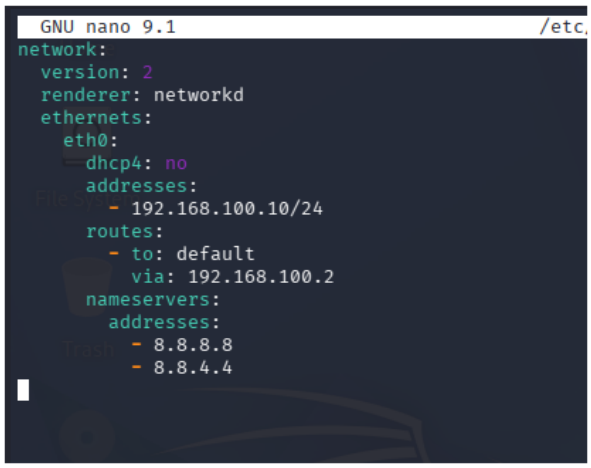
Step 2: Edit the netplan configuration file

```
sudo nano /etc/netplan/01-netcfg.yaml
```



Explanation: Opens the network configuration file for editing. (If the file doesn't exist, this command will create it.)

Step 3: Add the following content



Note: Replace eth0 with your actual interface name if it is different.

Explanation:

- dhcp4: no → Disables automatic IP
- addresses → Sets static IP
- gateway4 → Sets gateway
- nameservers → Sets DNS servers

Save the file: Ctrl + O → Enter → Ctrl + X

Step 4: Permissions + Apply the configuration

```
sudo chmod 600 /etc/netplan/01-netcfg.yaml
```



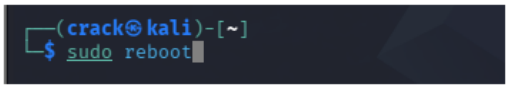

```
sudo netplan apply
```



Explanation: Applies the new static IP settings.

Step 5: Reboot your system

```
sudo reboot
```



Note: If facing problem in network setup (Troubleshooting)

Disable NetworkManager as we are using networkd & also disable Cloud-init

Cloud init

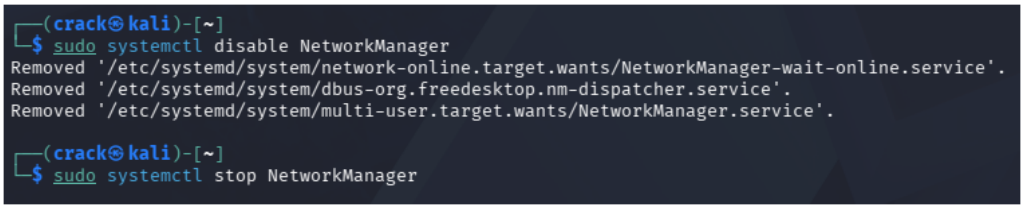
- `sudo mkdir -p /etc/cloud/cloud.cfg.d`
- `sudo tee /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg << EOF network: {config: disabled} EOF`



Network Manager

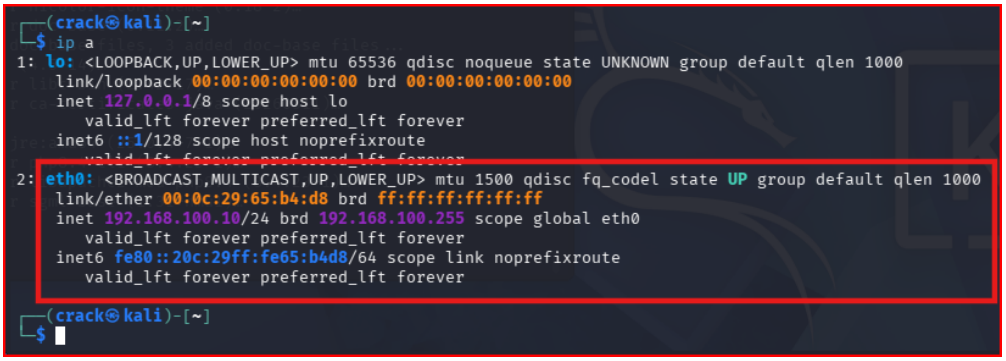
```
sudo systemctl disable NetworkManager
```

```
sudo systemctl stop NetworkManager
```



Step 6: Verify the IP

```
ip a
```



Explanation: Confirms that IP 192.168.100.10 is assigned.

Wazuh Agent setup on Kali Linux (Direct .deb Method)

Why Skip GPG Verification?

In modern Kali Linux:

- apt-key tool is deprecated (removed)
- Keyserver approach has compatibility issues
- For **isolated lab environments**, downloading directly from official Wazuh source is secure enough
- Direct .deb installation is faster and more reliable
- In production, GPG verification would be mandatory

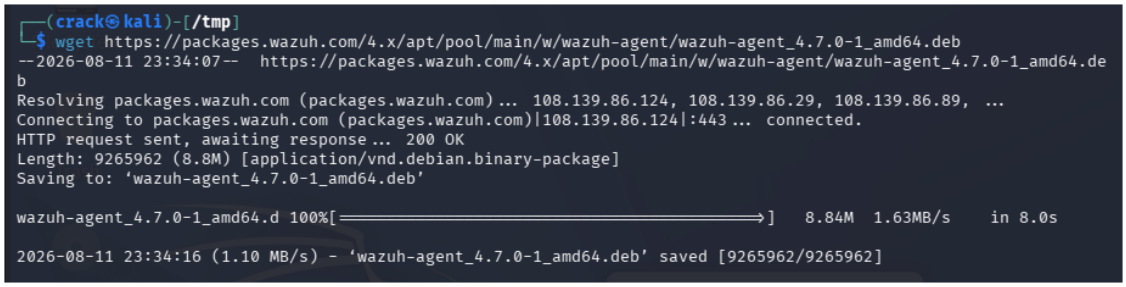
Change to Temporary Directory

cd /tmp



Download Wazuh Agent Package

wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.0-1_amd64.deb

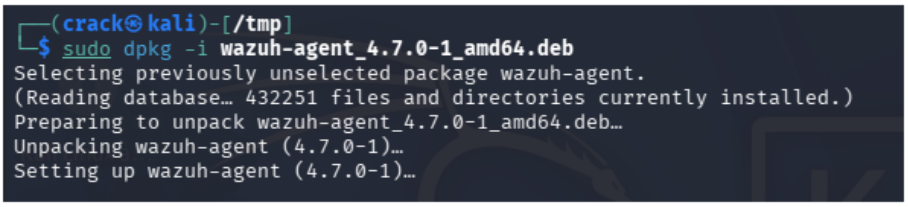


Explanation:

- Downloads official Wazuh agent .deb file (like an installer)
- File saved to /tmp/wazuh-agent_4.7.0-1_amd64.deb

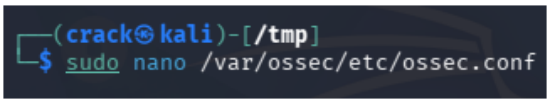
Install Package with dpkg

```
sudo dpkg -i wazuh-agent_4.7.0-1_amd64.deb
```



Open the configuration file

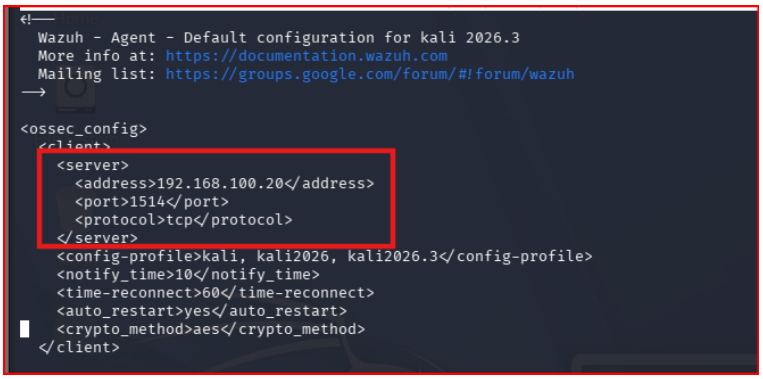
```
sudo nano /var/ossec/etc/ossec.conf
```



Reasoning: This is the main configuration file of the Wazuh agent. All monitoring rules, log sources, and manager connection settings are controlled from here.

Verify Manager Address

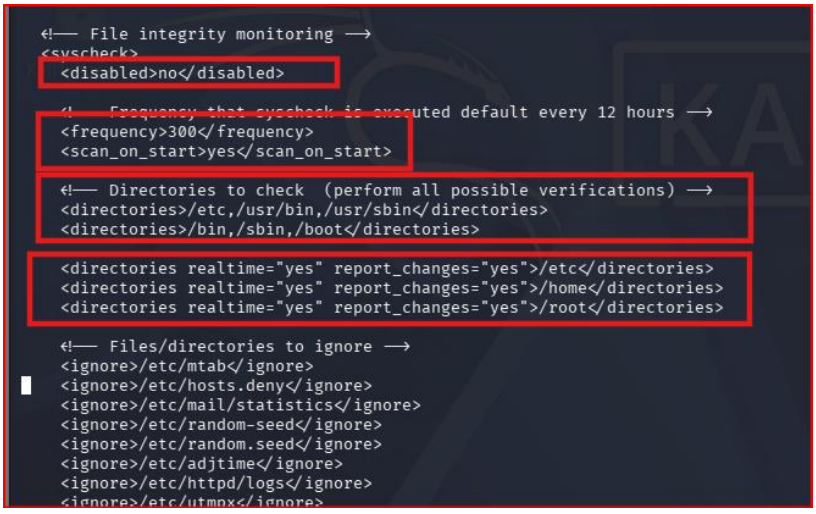
Make sure this block looks exactly like this:



Reasoning: This tells the agent where to send all collected logs and events. If this IP is wrong, the agent will never appear as Active in the dashboard.

Improve Syscheck Section

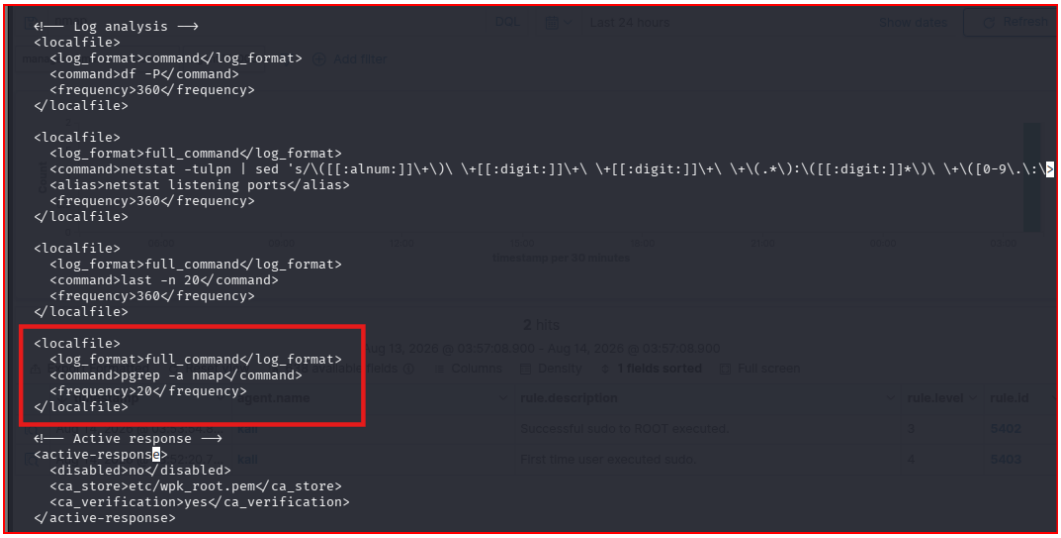
Your current realtime directories are good. Just make sure they look like this:



Reasoning:

- realtime="yes" → Detects file changes **immediately**
- report_changes="yes" → Shows what exactly changed in the file
- Frequency 300 is fine for lab testing (do **not** change it to 12 hours)

Command Monitoring (specifically Full Command Monitoring)



Reason: This makes the agent check every 20 seconds if the nmap process is running. When it finds it, it will generate a log that Wazuh can alert on.

Add Auth Log Monitoring (Most Important Missing Part)

Find the <localfile> section and add this block:

```
  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/messages</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/secure</location>
  </localfile>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/auth.log</location>
  </localfile>

</ossec_config>
```

Reasoning: Kali stores all authentication events (failed SSH, sudo, user logins) in /var/log/auth.log. Without this, many important security events will **not** appear in the Wazuh Dashboard.

Setting up rsyslog

Install and start rsyslog

```
sudo apt install rsyslog -y
```

```
┌──(crack㉿kali)-[~]
└─$ sudo apt install rsyslog -y
The following packages were automatically installed and are no longer required:
  bluez-hcidump  firmware-marvell-prestera  libntfs-3g89t64  libopenjph0.27  libvidstab1.1  libx265-215  network-manager-gnome
Use 'sudo apt autoremove' to remove them.

Installing:
  rsyslog

Installing dependencies:
  libestr0  libfastjson4  liblognorm5

Suggested packages:
  rsyslog-doc     rsyslog-mongodb        rsyslog-hiredis     rsyslog-docker      | rsyslog-gnutls
  rsyslog-mysql   rsyslog-elasticsearch  rsyslog-snmp        rsyslog-clickhouse  rsyslog-gssapi
  rsyslog-pgsql   rsyslog-kafka          rsyslog-kubernetes  rsyslog-openssl     rsyslog-relp

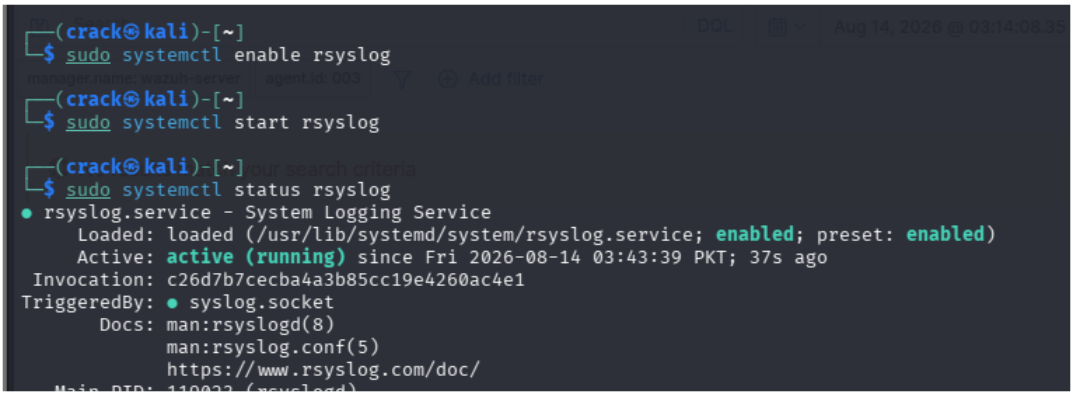
Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 2
  Download size: 985 kB
  Space needed: 2,636 kB / 21.4 GB available

Get:2 http://kali.download/kali kali-rolling/main amd64 libfastjson4 amd64 1.2304.0-3 [28.9 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 rsyslog amd64 8.2606.0-4 [873 kB]
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libestr0 amd64 0.1.11-2+b2 [9,596 B]
```

Start & Check status

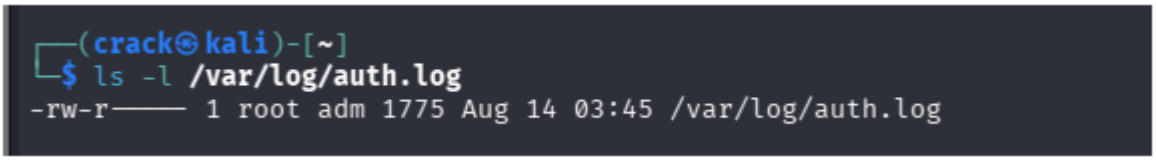
```
sudo systemctl enable rsyslog --now
```

```
sudo systemctl status rsyslog
```



Then verify the log file exists:

```
ls -l /var/log/auth.log
```



Reload systemd Daemon

```
sudo systemctl daemon-reload
```



Enable Auto-Start on Boot

```
sudo systemctl enable wazuh-agent
```



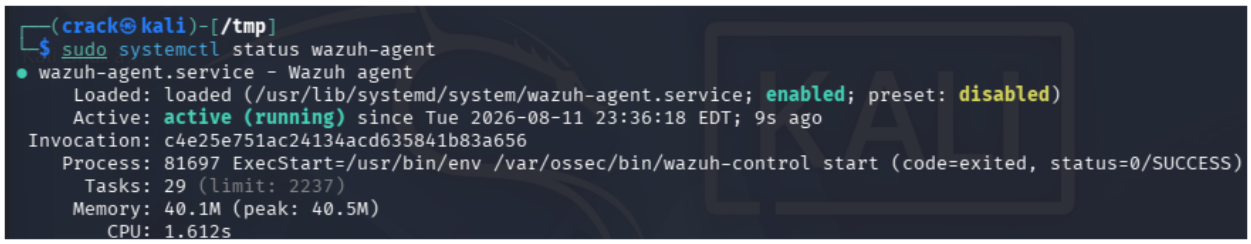
Start the Agent Service

```
sudo systemctl start wazuh-agent
```



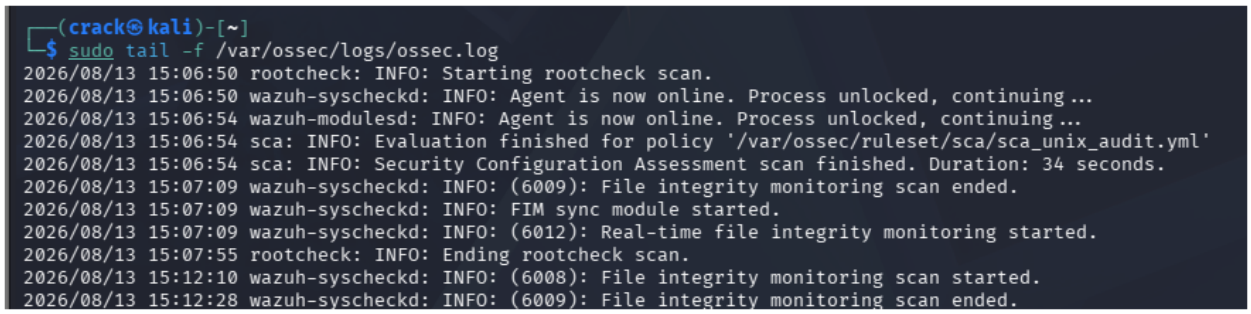
Verify Agent is Running

```
sudo systemctl status wazuh-agent
```



Verify Agent internal logs (Real Time Follow)

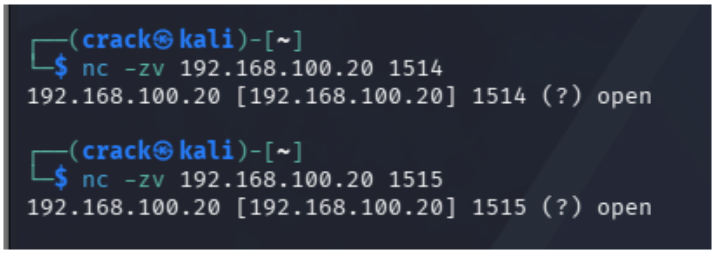
```
sudo tail -f /var/ossec/logs/ossec.log
```



Verify Agent manual connectivity test to Manager

```
nc -zv 192.168.100.20 1514
```

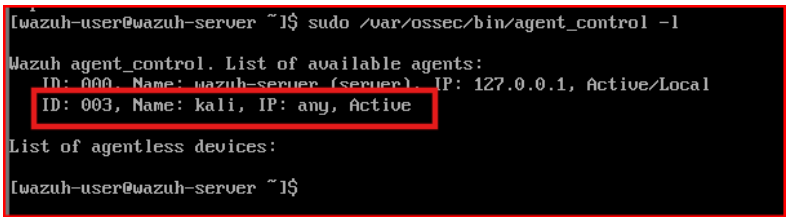
```
nc -zv 192.168.100.20 1515
```



Check connected agents

Machine: Wazuh OVA

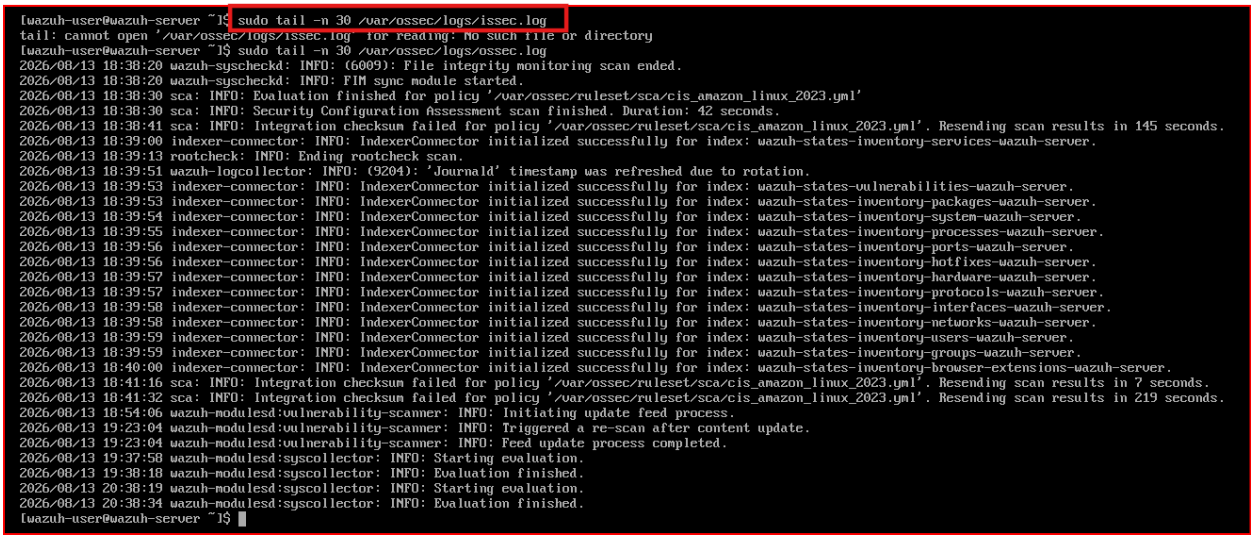
```
sudo /var/ossec/bin/agent_control -l
```



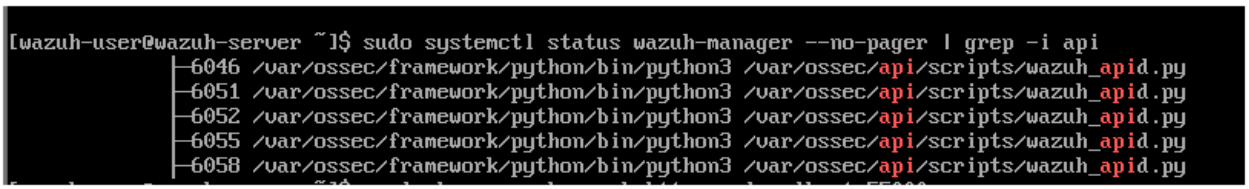
Expected: You should see your Kali agent (crack@kali or similar) as **Active**.

Check Manager logs

```
sudo tail -n 30 /var/ossec/logs/ossec.log
```



Check API status



Verify in Wazuh Dashboard

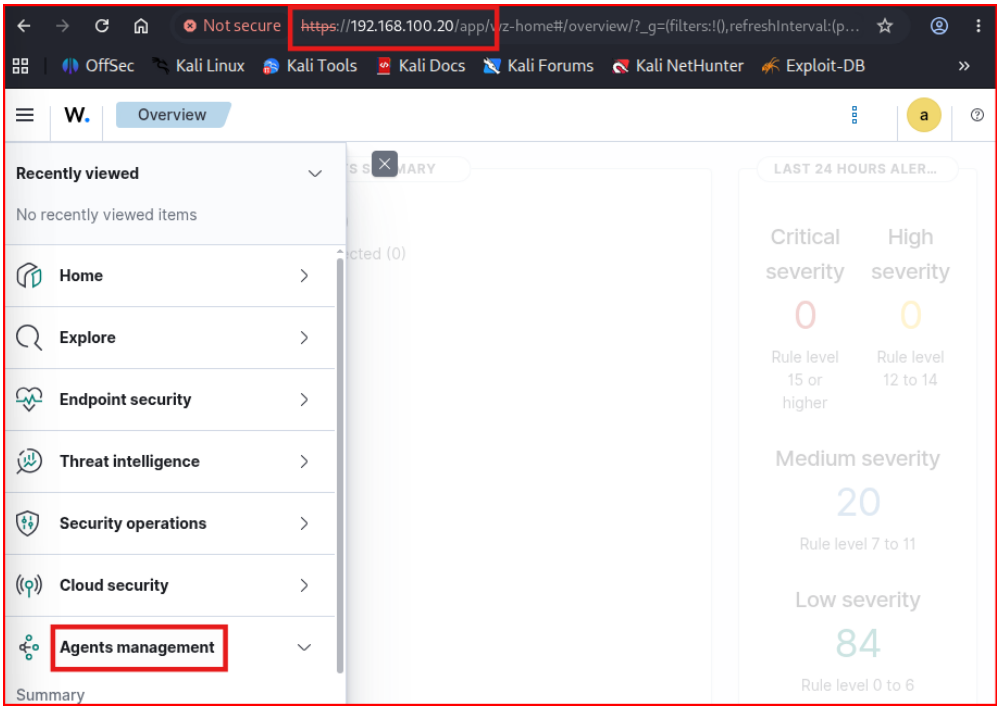
On your host machine or Kali, open browser:

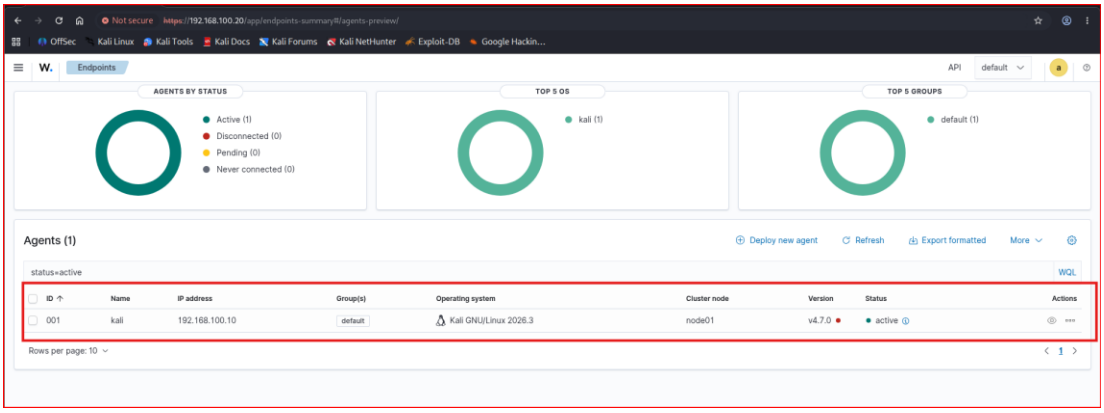
https://192.168.100.20

Login:

- Username: admin
- Password: admin (or changed password)

Check Agents:



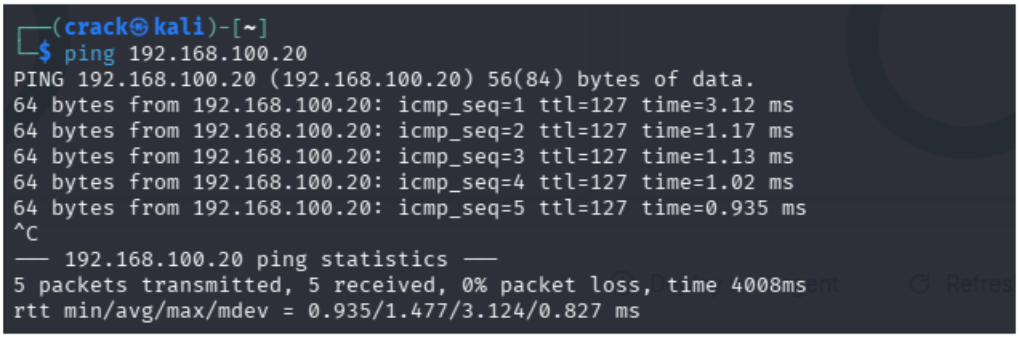


Testing & Validating

Test Connectivity Between VMs

From Kali Terminal:

```
ping -c 4 192.168.100.20
```

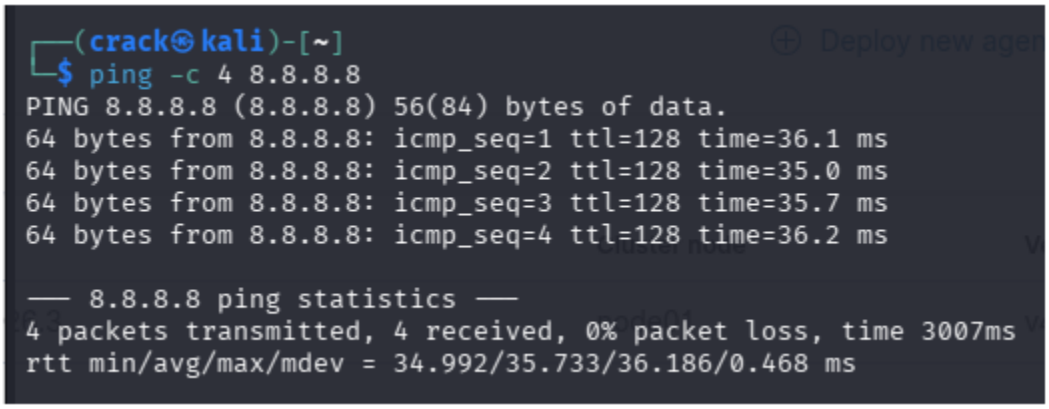


Explanation: Tests if Kali can reach Wazuh VM. Should see 4 replies with 0% packet loss.

Test Internet Access

From Kali:

```
ping -c 4 8.8.8.8
```

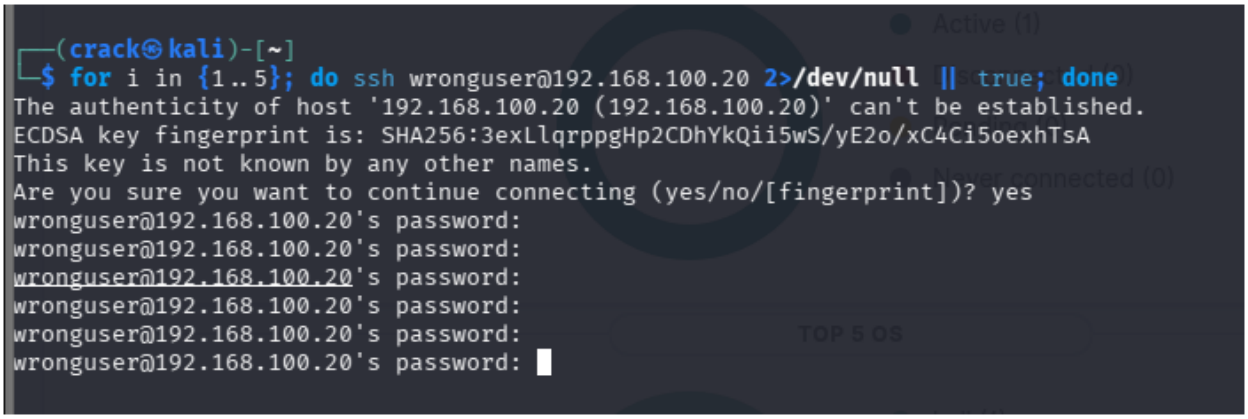


Explanation: Tests if lab has internet access (Google DNS). Should see 4 replies.

Generate Test Security Event 1 - Failed SSH Login

From Kali Terminal:

```
for i in {1..5}; do ssh wronguser@192.168.100.20 2>/dev/null || true; done
```



Explanation:

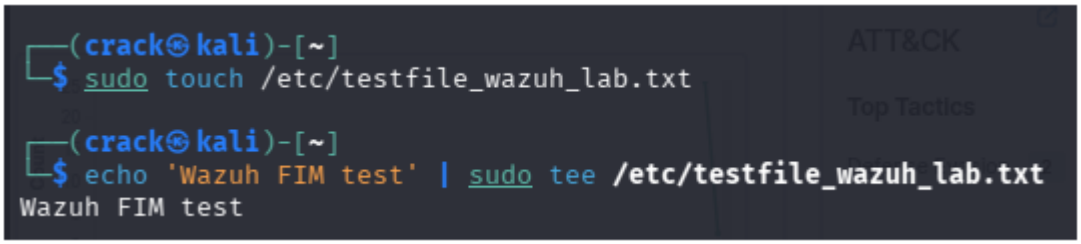
- Attempts 5 failed SSH logins to Wazuh VM
- Uses wrong username: wronguser
- Wazuh will detect this as brute force attempt
- Creates security events in Wazuh Dashboard

Generate Test Security Event 2 - File Integrity Monitoring

From Kali Terminal:

```
sudo touch /etc/testfile_wazuh_lab.txt
```

```
echo 'Wazuh FIM test' | sudo tee /etc/testfile_wazuh_lab.txt
```



Explanation:

- Creates a new file in /etc directory
- Wazuh monitors /etc by default (File Integrity Monitoring)
- Wazuh detects file creation event
- Should appear in dashboard

Integrity Test 2

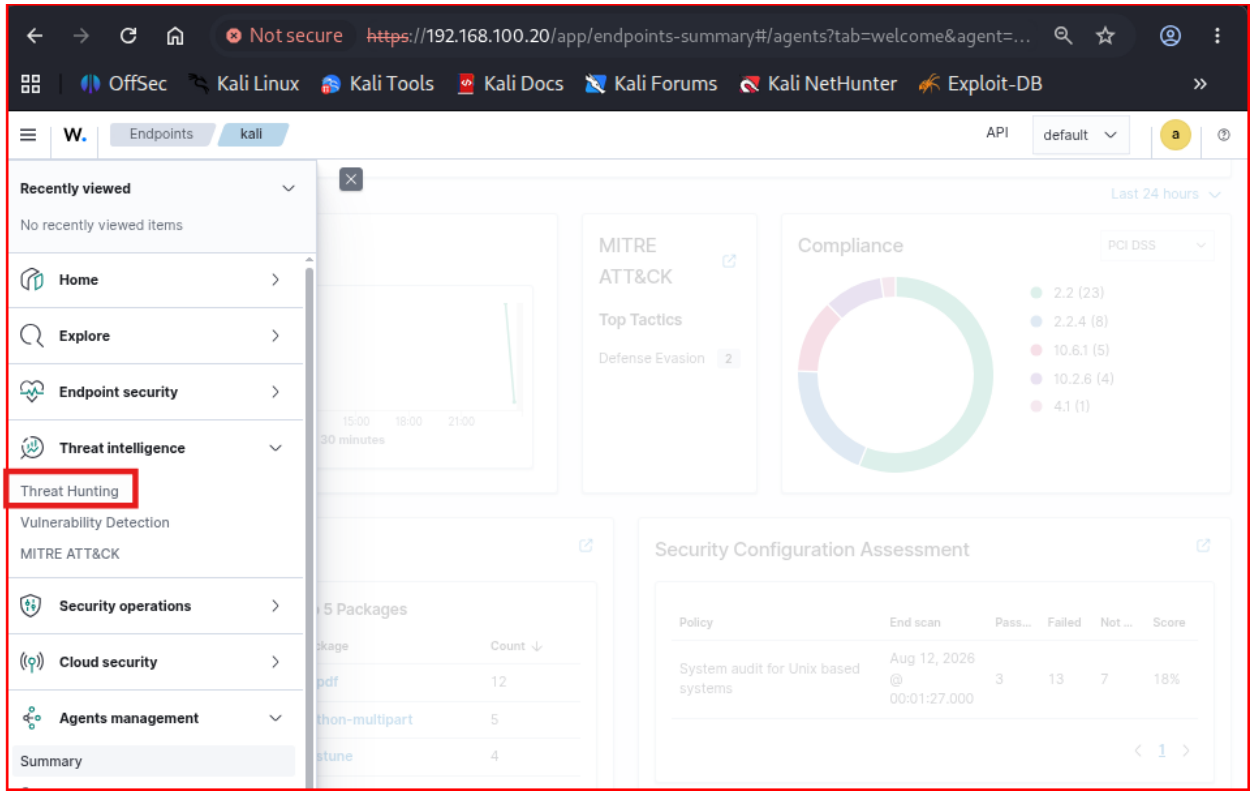
```
┌──(crack㉿kali)-[~]
└─$ sudo touch /etc/test_wazuh_fim_$(date +%s).txt
[sudo] password for crack:
```

Check Security Events in Wazuh Dashboard

On your host machine or Kali, open browser:

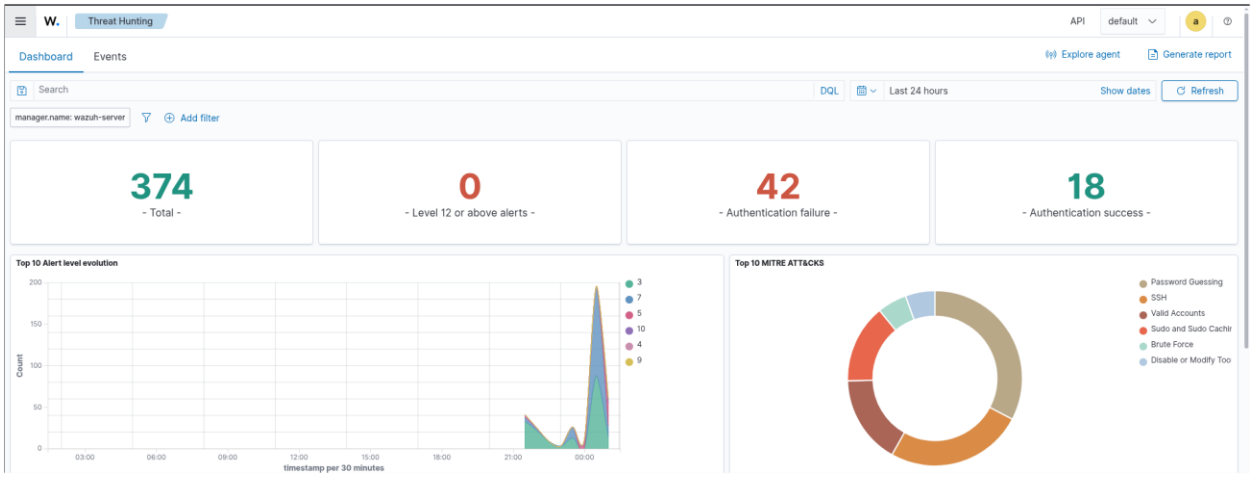
Navigate to Security Events:

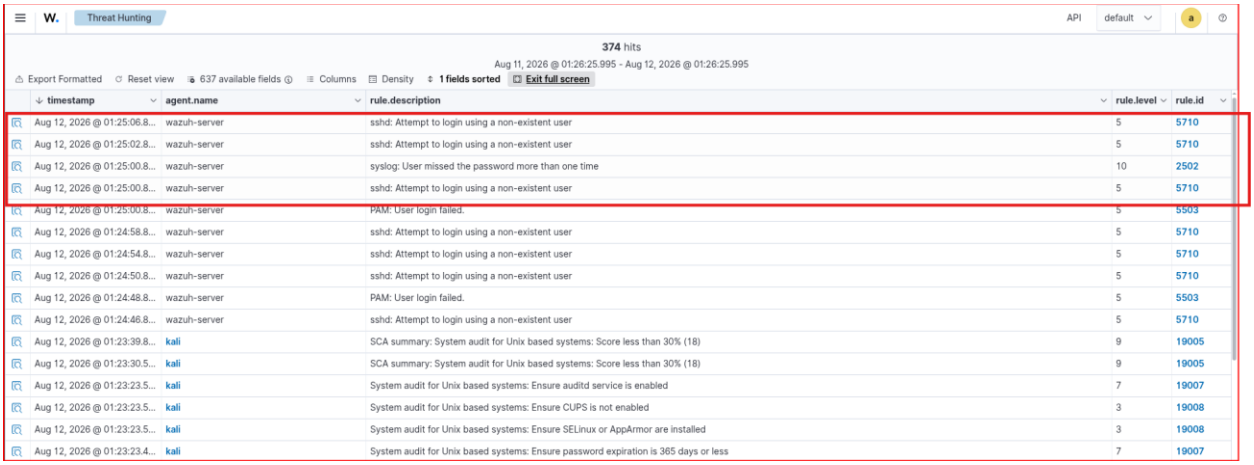
1. Click **Threat Hunting** (left sidebar)



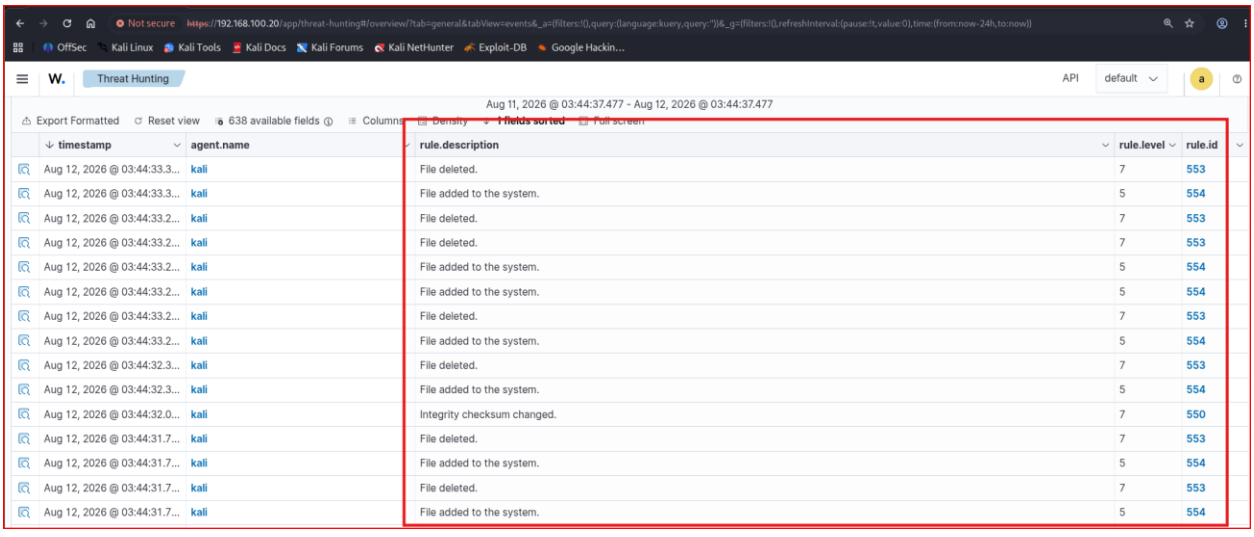
Security Events Checkup:

Authentication failure (from SSH attempts)





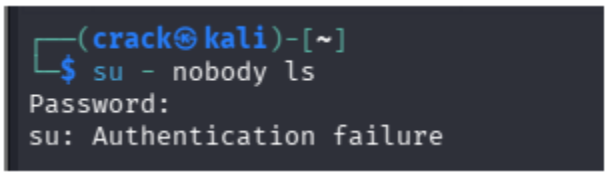
File integrity alerts checking



Performing Multiple Test & Verifying the results

Test Authentication Failure

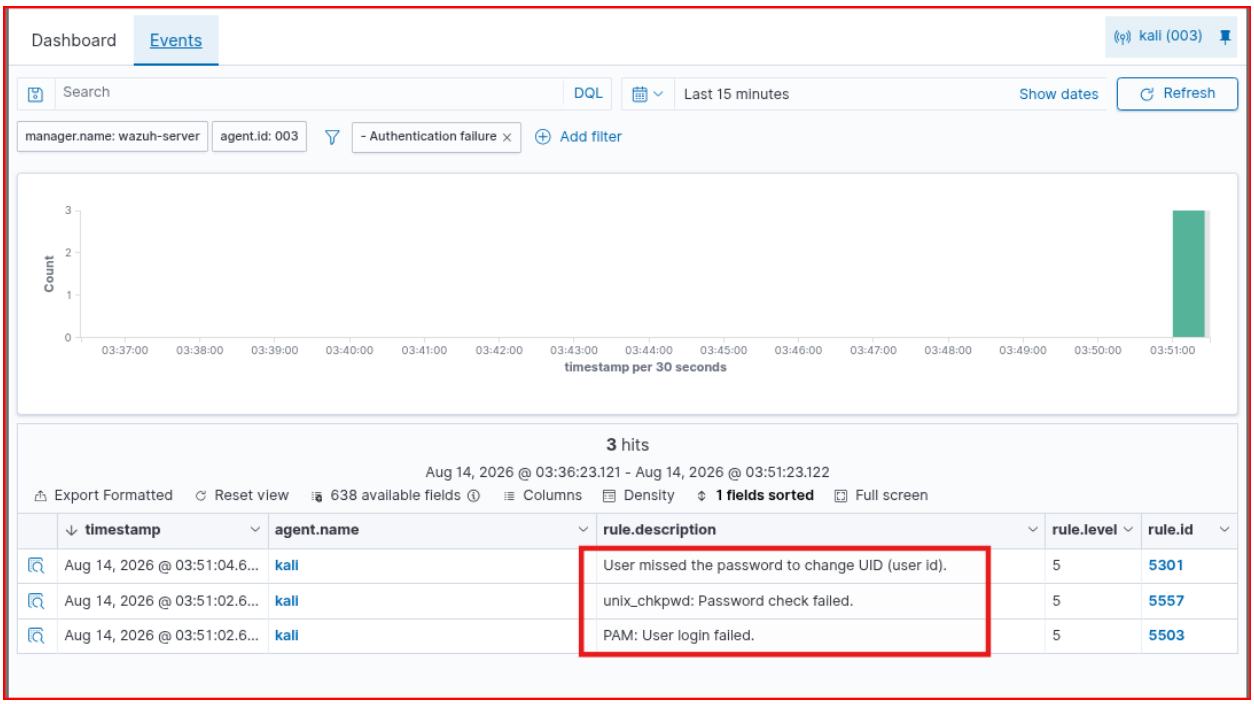
```
su - nobody
```



You can clearly see in the last screenshot:

- Rule 5503 → PAM: User login failed
- Rule 5557 → unix_chkpwd: Password check failed
- Rule 5301 → User missed the password to change UID

This means the rsyslog + /var/log/auth.log setup is successful.

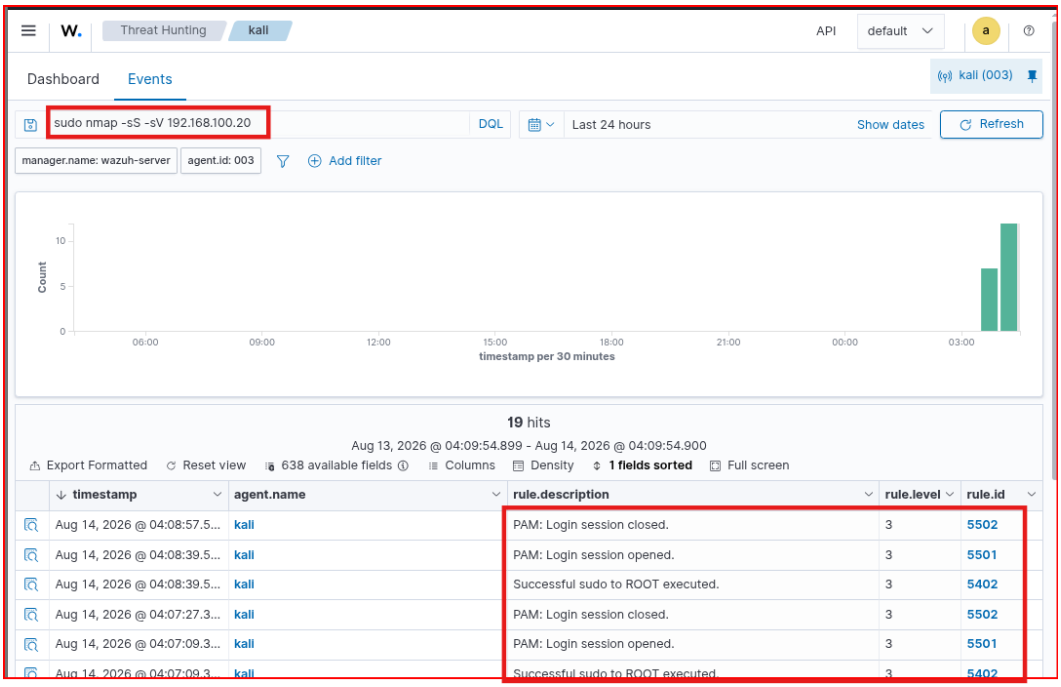


Testing & Verification – Nmap Port Scan

```
┌──(crack㉿kali)-[~]
└─$ sudo nmap -sS -sV  192.168.100.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-14 04:08 +0500
Nmap scan report for 192.168.100.20 (192.168.100.20)
Host is up (0.00070s latency).
Not shown: 998 closed tcp ports (reset)
PORT    STATE SERVICE   VERSION
22/tcp  open  ssh       OpenSSH 8.7 (protocol 2.0)
443/tcp open  ssl/https
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://
nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port443-TCP:V=7.99%T=SSL%I=7%D=8/14%Time=6A7E4E83%P=x86_64-pc-linux-gnu
SF:%r(GetRequest,15C,"HTTP/1\.1\x20302\x20Found\r\nlocation:\x20/app/login
SF:\?\r\nosd-name:\x20wazuh-server\r\nx-frame-options:\x20sameorigin\r\nca
SF:che-control:\x20private,\x20no-cache,\x20no-store,\x20must-revalidate\r
SF:\nset-cookie:\x20security_authentication=;\x20Max-Age=0;\x20Expires=Thu
SF:,\x2001\x20Jan\x201970\x2000:00:00\x20GMT;\x20Secure;\x20HttpOnly;\x20P
SF:ath=/\r\ncontent-length:\x200\r\nDate:\x20Thu,\x2013\x20Aug\x202026\x20
SF:23:08:51\x20GMT\r\nConnection:\x20close\r\n\r\n")%r(HTTPOptions,143,"HT
SF:TP/1\.1\x20404\x20Not\x20Found\r\nosd-name:\x20wazuh-server\r\nx-frame-
SF:options:\x20sameorigin\r\ncontent-type:\x20application/json;\x20charset
SF:=utf-8\r\ncache-control:\x20private,\x20no-cache,\x20no-store,\x20must-
SF:revalidate\r\ncontent-length:\x2060\r\nDate:\x20Thu,\x2013\x20Aug\x2020
SF:26\x2023:08:51\x20GMT\r\nConnection:\x20close\r\n\r\n{\"statusCode\":40
SF:4,\"error\":\"Not\x20Found\",\"message\":\"Not\x20Found\"}")%r(FourOhFo
SF:urRequest,1C9,"HTTP/1\.1\x20401\x20Unauthorized\r\nosd-name:\x20wazuh-s
SF:erver\r\nx-frame-options:\x20sameorigin\r\ncontent-type:\x20application
SF:/json;\x20charset=utf-8\r\ncache-control:\x20private,\x20no-cache,\x20n
```

Result on Kali:

- Nmap successfully scanned the Wazuh Manager (192.168.100.20)
- Open ports detected: 22 (SSH) and 443 (HTTPS)



Result on Wazuh Dashboard (Agent: kali):

When searching for the exact command sudo nmap -sS -sV 192.168.100.20, the following related events were generated:

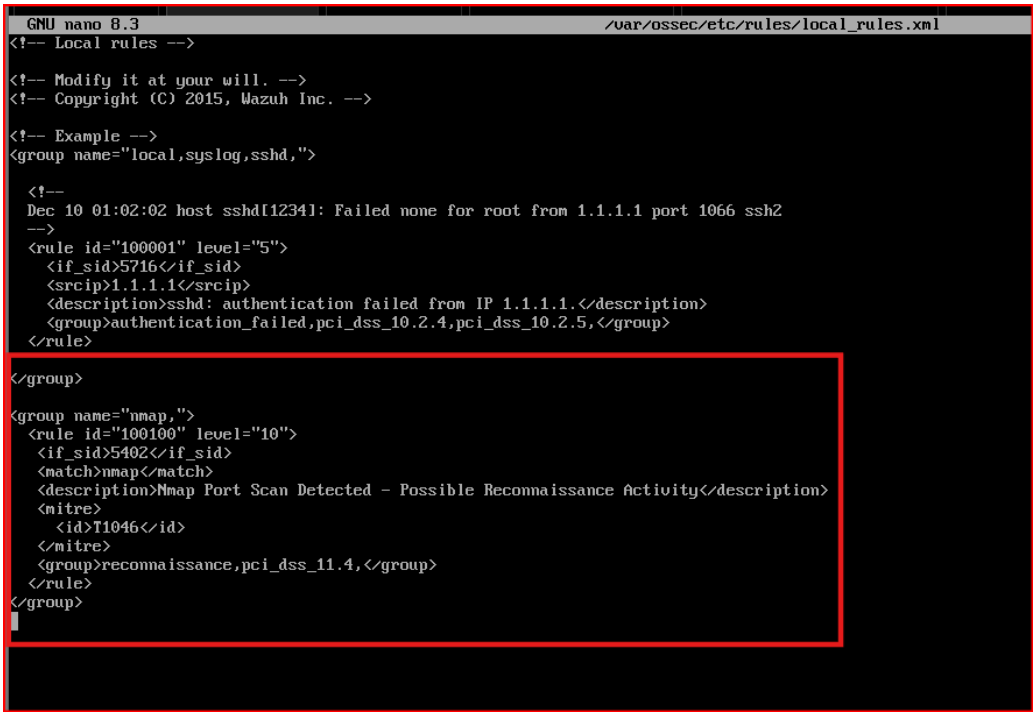
Create Custom Rule for Nmap (Wazuh Manager (OVA))

1. Open the custom rules file on the Wazuh Manager

```
sudo nano /var/ossec/etc/rules/local_rules.xml
```



2. Add the following rule at the end of the file (before </group> if it exists, or just at the bottom):



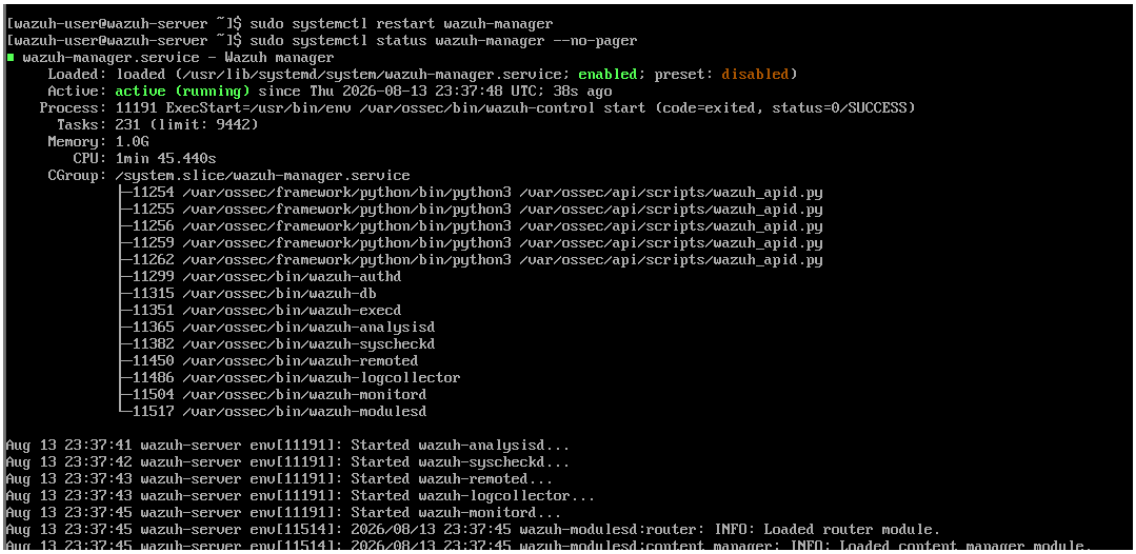
Explanation of the rule:

- id="100100" → Custom rule ID (always use 100000+ range)
- level="10" → High severity so it stands out
- if_sid>5402 → This rule will trigger when a "Successful sudo to ROOT" event contains the word "nmap"
- match>nmap → Looks for the word nmap in the log
- MITRE T1046 = Network Service Discovery

Restart Wazuh Manager

```
sudo systemctl restart wazuh-manager
```

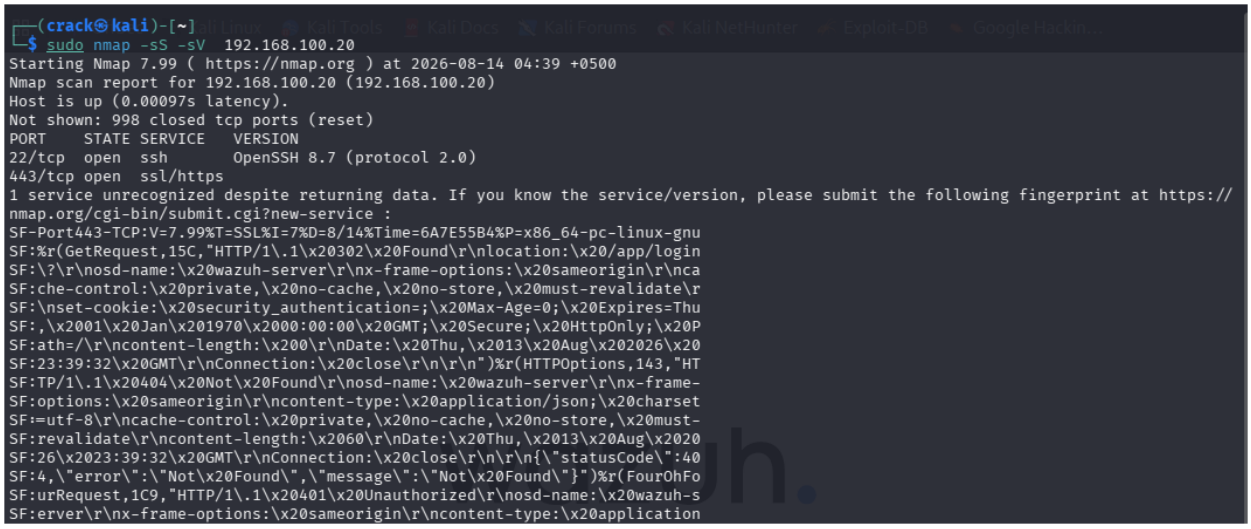
```
sudo systemctl status wazuh-manager --no-pager
```



Testing Custom Rule

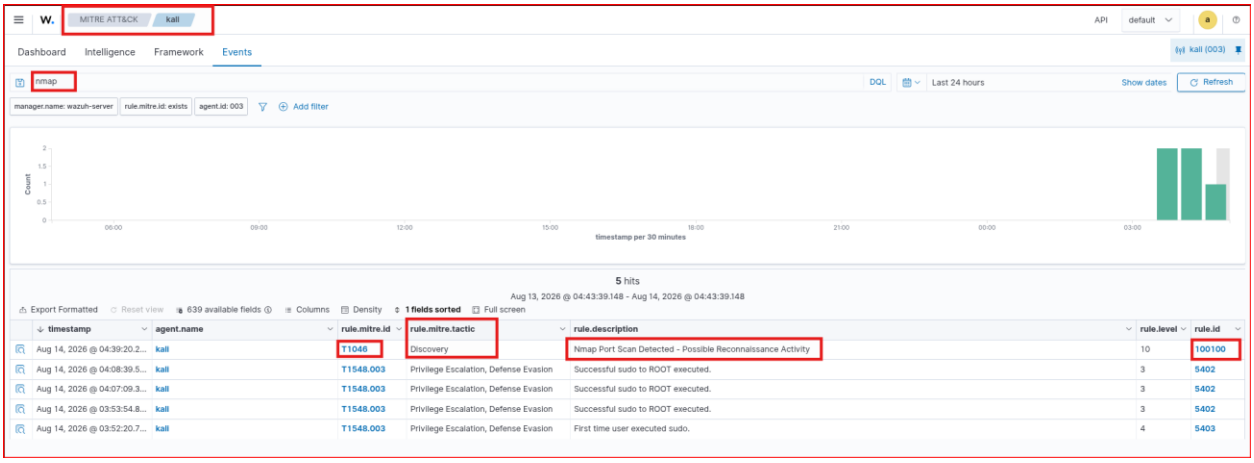
Test from Kali

```
sudo nmap -sS -sV 192.168.100.20
```



Check Dashboard

- Go to **Threat Hunting → Events**
- Filter by agent **Kali**
- Look for **Rule ID 100100**



Results:

The custom rule is working perfectly.

- **Rule ID:** 100100
- **Description:** Nmap Port Scan Detected - Possible Reconnaissance Activity
- **Level:** 10
- **Agent:** kali

Conclusion

The project successfully demonstrated that Wazuh can provide effective **centralized security monitoring and threat detection** in a controlled environment. The configured agent successfully forwarded security events, while Wazuh detected authentication failures, file changes, and network scanning activity.

The successful Nmap test and custom **Level 10 detection rule (100100)** further demonstrated that Wazuh can be customized to identify specific reconnaissance activity and associate it with **MITRE ATT&CK T1046**.

In conclusion, the lab successfully achieved its objective of building, testing, and validating a functional SIEM-based security monitoring system.